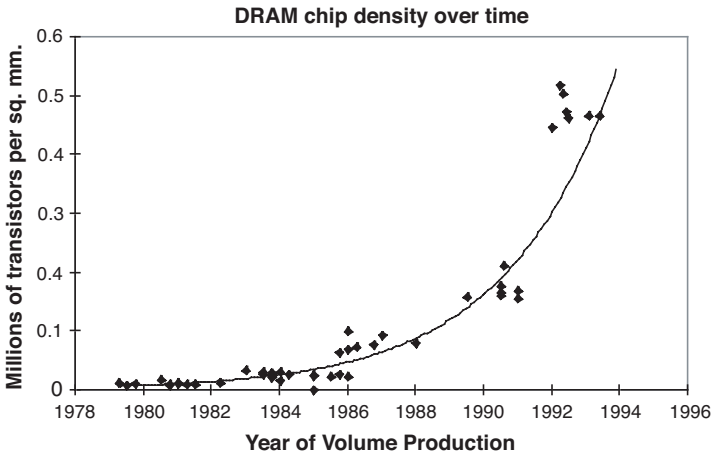


Figure 5. Transistors per mm² vs. year of volume production in DRAM.[22]

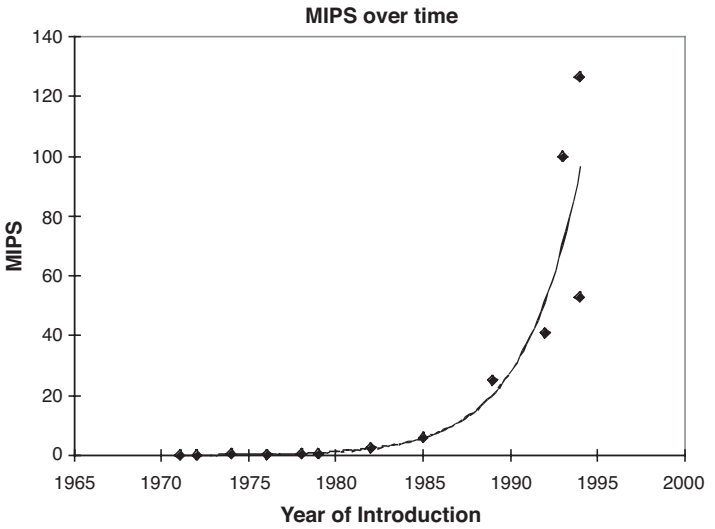


Figure 6. Processor speed in MIPS vs. year of introduction.

processors for his data points. This significantly increased the law's simplicity, especially because DRAMs are based on a stable technology. Technical innovation was therefore not a factor in increasing DRAM complexity, and technical cleverness had already ceased to be an issue since the failure of CCDs. That meant that DRAM progress was solely driven by increases in line density and die size, both of which grew at a fairly constant pace.

The standardization around DRAMs had the added advantage of allowing the pace of the law in the VLSI era to be more easily measured. In earlier periods, differing logic families and chip types made it difficult to compare ICs, a problem that does not affect DRAMs in the VLSI era. As a result, it is possible to directly calculate the accuracy of Moore's law for DRAM (see Figure 5).

Forty-three points, representing the transistor density of DRAM chips in transistors per square millimeter, were graphed. These points included chips from five generations of DRAM, with dates of commercial introduction from the late 1970s until the mid-1990s. The points fell into an exponential curve that approximated a 31 percent improvement per year, or a doubling around every 26 months. The curve fit the data fairly well, with "Moore's law," as characterized by this 26-month cycle, accounting for 76 percent of the variation in the data points. This implies that, at least through much of the VLSI era, the 26-month cycle called Moore's law is an accurate indicator of technical progress.[23]

In addition, with the increasing importance of microprocessors, they, too, were incorporated into the VLSI conception of Moore's law. The emphasis on microprocessors also changed the nature of the Moore's law measurement. For the first time, the law was used to refer to trends other than increasing complexity. This was because microprocessors are not as transistor-dense as DRAM chips, so they could not be measured on the same scale as memory chips. Instead, they are usually measured by either the total number of transistors on a chip, or else by their speed of operation, measured in the number of millions of instructions per second (MIPS) that a processor can carry out.

Typically, between microprocessor generations, a company will release a series of faster, improved microprocessors based on the same design. An example would be Intel's original 486 chip, a 20-MIPS model released in 1989, which was followed by an improved 27-MIPS chip in 1990, and a 41-MIPS chip in 1991. All these had the same transistor count as the first 486 chip, but achieved better performance thanks to a variety of design improvements. For this reason, microprocessors are often graphed by performance, in MIPS, instead of by transistor count (see Figure 6).

The graph in Figure 6, too, indicates a curve that approximates the VLSI Moore's law exponential, with 30 percent improvement per year, and a fit of 97 percent. The graph does seem to indicate two diverging sets of points, however. The lower two points represent improved versions of older chip models, sold as the basis for "economy" computers. When they are removed, the fit is much better.

These three measurements—DRAM complexity, processor transistor count, and processor speed—are all considered part of Moore's law in the VLSI era, which indicates that the law's current form states that chip performance